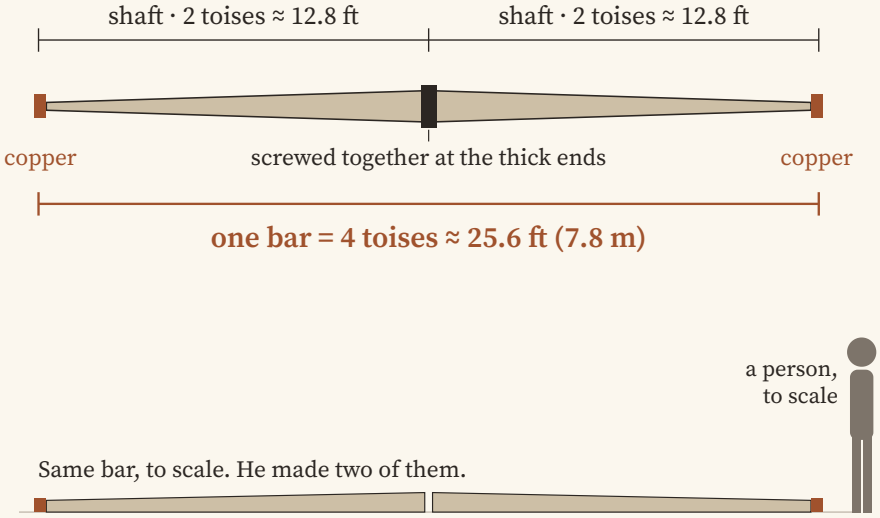
How Jean Picard measured a straight line 6.9 miles long

Before he could measure the Earth, he needed one distance on the ground he could trust. He chose a straight paved road south of Paris and measured it with wooden rods.

1 The rods

Four straight wooden pike shafts, each 2 toises (≈ 12.8 ft) long. He screwed them together in pairs at the thick end. That made two measuring bars, each 4 toises (≈ 25.6 ft) long. Copper plates on the ends let one bar butt cleanly against the next.



2 The road and the cord

The road ran straight from the middle of the Villejuif windmill to the nearest corner of the Juvisy pavilion. A long cord was pulled tight along it. Each time the cord was moved, it was aimed again at both ends.



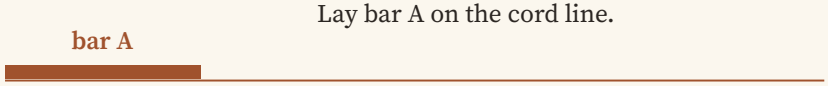
Unit key 1 Paris toise ≈ 6.4 ft (≈ 1.95 m) · 1 toise = 6 Paris feet · 1 Paris foot ≈ 1.07 ft

Source: Jean Picard, *Mesure de la Terre* (1671). 1 Paris toise ≈ 6.4 ft (≈ 1.95 m).

3 Leapfrog the two bars

Lay one bar down. Set the other end to end against it, along the cord. Pick up the first and carry it ahead. Repeat, always on the same line.

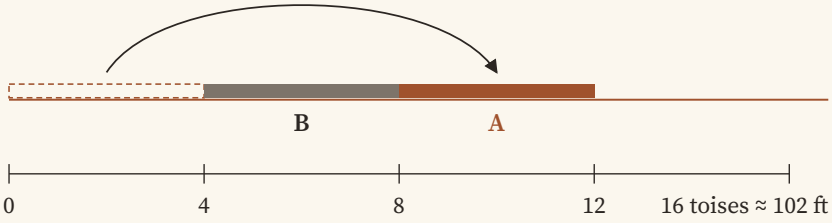
1 Lay bar A on the cord line.



2 Butt bar B against the end of A.



3 Lift A, carry it ahead, butt it against B.

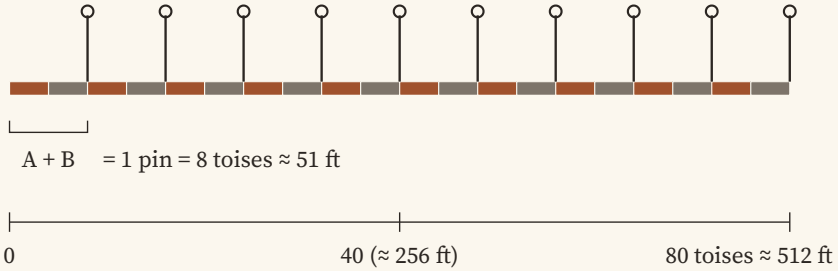


Each tick = 1 bar = 4 toises ≈ 25.6 ft.
One bar always stays on the ground, so the line never loses its place.

4 Counting with pins

The man at the head of the bars carried 10 iron pins. He left one on the ground each time he set his bar down. His bar went down every other time, so each pin stood for two bars: 8 toises (≈ 51 ft).

One round: 20 bars, 10 pins



10 pins left on the ground = 80 toises ≈ 512 ft

Write down 80 toises (≈ 512 ft).
Pick the pins up and start the next round.

For the whole road (simple arithmetic)

5,663 toises (≈ 6.9 mi) ÷ 8 ≈ 708 pins dropped

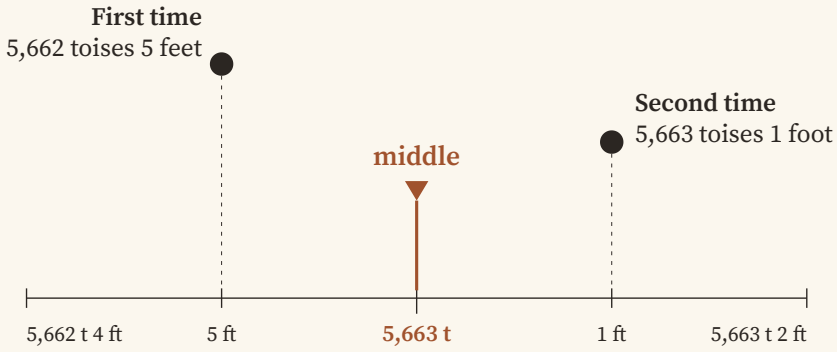
5,663 toises ÷ 80 ≈ 71 rounds of 10 pins

Counting by pins means no one has to keep 700 numbers in his head.

Source: Jean Picard, *Mesure de la Terre* (1671): “chaque fiche valait huit toises … on marquait 80 toises.”

5 Measure it twice, take the middle

He measured the whole road two separate times. The two answers differed by just 2 Paris feet (≈ 2.1 ft). He took the middle value.



Accepted length: 5,663 toises ≈ 6.9 miles (11.04 km)

Gap: 2 Paris feet ≈ 2.1 ft (65 cm) over 6.9 miles
≈ 1 part in 17,000

Horizontal scale greatly stretched: whole axis = 4 Paris feet ≈ 4.3 ft. 1 toise = 6 Paris feet.

6 A second base as a check

At the end, he checked all his work against a second baseline of 3,902 toises (≈ 4.7 miles), measured the same way.



Drawn to the same scale.

Same bars, same cord, same pins.
If the two agree, the work in between can be trusted.

What the sources do not say. The sources found mention no correction for temperature (wood swelling or shrinking). They do not describe a slope correction for this road either.

Units. 1 Paris toise ≈ 6.4 ft (≈ 1.95 m) = 6 Paris feet; 1 Paris foot ≈ 1.07 ft. Feet and mile values are modern conversions; counts of bars, pins and rounds are simple arithmetic.

Source. Jean Picard, *Mesure de la Terre* (Paris, 1671), as reproduced at ertia2.free.fr and the 1780 abridgement at assprouen.free.fr.